

Physiology & Biochemistry

Paper –2: Physiology and Biochemistry

Departmental Objectives of Physiology

At the end of the course in physiology the BDS students will be able to:

- demonstrate basic knowledge and understanding on the normal functions of all the organ systems of human body and apply it as a background for clinical subjects.
- interpret normal function with a view to differentiate from abnormal function.
- perform physiological experiments.
- interpret experimental and investigation data
- develop sound attitude for continuing self-education to improve efficiency & skill in physiology.

Departmental Objectives of Biochemistry

At the end of the course in biochemistry the students will be able to:

- demonstrate basic knowledge and understanding on major biomolecules, enzymes, hormones and nutrients and of fundamental chemical principles involved in body mechanism upon which life process depends
- demonstrate skills in performing and interpreting common Bio-chemistry laboratory tests and procedures with emphasis on those used in Bangladesh.
- develop sound attitude towards the need for continuing self education

List of Competencies to acquire in Physiology :

In the process of completing these courses, students will acquire the following competencies:

- Describe transport processes across the plasma membrane, resting membrane potential, action potentials.
- Explain muscle contraction and relaxation.
- Describe the function of heart and circulatory system
- Describe respiratory processes with the knowledge of structures, ventilation, diffusion, blood flow, gas transport, mechanics of breathing, and control of ventilation.
- Explain the mechanism of work of the brain at the neuronal systems level.
- Describe the autonomic nervous system and special sense
- Describe the function of endocrine physiology
- Describe human reproduction, functional changes in the reproductive tract, the formation of sperm & ovum.
- Demonstrate adequate knowledge and develop skill for performing physiology laboratory tests and interpreting these normal functions with a view to differentiate from abnormal conditions. such as:
- Measurement of blood pressure
- Examination of radial pulse.
- Estimation of Hb concentration.
- Determination of differential count of white blood cell (WBC).
- Determination of bleeding time & clotting time.
- Determination of blood grouping & cross matching.

- Determination of erythrocyte sedimentation rate (ESR).
- Recording of body temperature.
- Elicitation of light reflex.

List of Competencies to acquire in Biochemistry:

After completing the course of biochemistry in BDS course the students will be able to-

- apply the knowledge and understanding of biochemistry in medicine and surgery.
- demonstrate knowledge and understanding of the biomolecules forming the structure of the human body, their functions and their role in health and diseases.
- explain the role of enzymes in the diagnosis of diseases.
- explain metabolic reactions in the body.
- explain the role of liver in metabolism
- explain the mechanism of maintenance of homeostasis by regulating both the composition and volume of ECF compartment by the kidney
- describe the water and electrolyte content of human body and their functions
- describe the types, causes and consequences of dehydration and over hydration.
- explain the causes and the consequences of electrolyte imbalance.
- describe the sources of acids and bases in our body and the mechanism of their normal balance. Explain the causes and consequences of acidosis and alkalosis and the parameters to diagnose them.
- demonstrate knowledge and understanding about nutrients, balanced diet.
- explain the basis of genetics and molecular biology.
- diagnose diabetes mellitus, impairment of renal, liver and thyroid functions.

Organization of the Course

The course is offered in 2 terms (1st & 2nd) and total 1&1/2 years for phase –1 BDS Course.

Distribution of teaching /learning hours

Lecture	Tutorial	Practical	Demonstration +Dissection +Card exam	Total	Integrated teaching (Common)	Formative Exam		Summative exam	
						Preparatory leave +post-term leave	Exam time	Preparatory leave	Exam time
132 hrs	124 hrs	60 hrs	-	316 hrs	10 hrs	20 days	42 days	30 days	30 days

Professional Examination:

Marks distribution of Assessment of Physiology & Biochemistry

(Total marks 300)

- Written = 100 (SAQ=70 + MCQ =20 + Formative Assessment =10)
- SOE = 100
- Practical = 100

Teaching/learning methods, teaching aids and evaluation

Teaching Methods			Teaching aids	In course evaluation
Large group	Small group	Self learning		
Lecture, Integrated teaching	Tutorial Practical Demonstration	Assignment, self assessment & self study.	Computer & Multimedia Chalk & board White board & markers OHP Slide projector Flip Chart Models Specimens projector Study guide & manuals.	Item examination(oral) Practical item examination(Oral & practical) Card completion Examination (Written) Term final Examination(Written, oral & practical)

Related Equipments: Microscope, photoelectric colorimeter, test tube, glass slide, centrifuge machine, micro pipette, chemicals & reagents, Sphygmomanometer, Stethoscope, Clinical thermometer, Spirit, Pencil torch, models, specimens, Haemocytometer, Shahlis Haemometer, Haematocrit tube, Westerngren ESR tube & ESR Stand etc.

General Physiology

Learning Objectives	Contents	Teaching Hours
At the end of the course the students will be able to explain about : ● Goal of physiology. ● Principles of homeostasis and control system ● Functional organization of the human body & cell physiology. ● Cell membrane transport. ● Membrane potential, resting membrane potential and action potential. ● Muscles, contraction and relaxation of skeletal muscle. ● Neuromuscular transmission.	CORE: ● Goal of Physiology ● Homeostasis: definition, basic concept and feed back mechanism. ● The cell: cell organization, cell membrane, cell organelles and their functions. ● Transport processes across the cell membrane ● Membrane potential, resting membrane potential and action potential. ● Contraction and relaxation of skeletal muscle. ● Neuromuscular transmission.	L=5 hrs T=6 IT=1

Blood

Learning Objectives	Contents	Teaching Hours
At the end of the course the students will be able to: ● Describe the composition & functions of blood. ● Demonstrate knowledge about plasma proteins. ● Demonstrate knowledge about the formation , morphology, types & functions of RBC,WBC & platelets. ● Describe the types, function and fate of haemoglobin. ● Demonstrate knowledge about blood grouping & blood transfusion. ● Describe about hemostasis & coagulation. ● Describe about the bleeding disorders.	CORE: ● Blood: composition & functions. ● Plasma proteins: normal values, properties, functions & effect of hypoproteinaemia ● Development and normal values of formed elements. ● RBC: morphology, function and erythropoiesis. ● Hemoglobin: types, functions & fate of hemoglobin. ● Red blood cell indices : PCV, MCV, MCH & MCHC ● Anaemia, Polycythemia & Jaundice: definition & classification. ● WBC: Classification, morphology, properties & functions, leucocytosis , leucopenia, Platelet: morphology & functions. ● Hemostasis: definition & events. ● Coagulation: definition, clotting factors & mechanism ● Blood grouping: ABO & Rh system ● Hazards of blood transfusion & Rh incompatibility. Additional/Applied Physiology ● Bleeding disorder: thrombocytopenic purpura & hemophilia, tests for bleeding disorder.	L=15 T=12 P=20

Blood

Learning Objectives	Contents	Teaching Hours
At the end of the course the students will be able to : ● describe the properties of cardiac muscle ● describe the origin and conduction of cardiac impulse. ● demonstrate knowledge about events of cardiac cycle. ● interpret the heart sounds. ● define and interpret a normal ECG. ● describe local & humoral control of blood flow ● describe cardiac output, venous return & peripheral resistance. ● explain about the heart rate & radial pulse. ● describe the blood pressure and its regulation.	CORE : ● Cardiac muscle: physiological anatomy, properties. ● Junctional tissues of the heart: generation of cardiac impulse & its conduction. ● Cardiac cycle: events, pressure & volume changes during different phases ● Heart sounds: types & characteristics ● ECG: definition & interpretations . ● local & humoral control of blood flow ● Cardiac output : definition and factors affecting cardiac output. ● Venous return & Peripheral resistance: definition & factors affecting.. ● Heart rate: definition, normal rate & regulation. ● Radial pulse: definition & characteristics. ● Blood pressure: definition, types, measurement & regulation of arterial blood pressure. Additional /Applied Physiology Cardiac arrhythmias: tachycardia, bradycardia & heart block	L=15 T=12 P=10 IT=01

Cardiovascular System

Learning Objectives	Contents	Teaching Hours
At the end of the course the students will be able to : ● describe the properties of cardiac muscle ● describe the origin and conduction of cardiac impulse. ● demonstrate knowledge about events of cardiac cycle. ● interpret the heart sounds. ● define and interpret a normal ECG. ● describe local & humoral control of blood flow ● describe cardiac output, venous return & peripheral resistance. ● explain about the heart rate & radial pulse. ● describe the blood pressure and its regulation.	CORE : ● Cardiac muscle: physiological anatomy, properties. ● Junctional tissues of the heart: generation of cardiac impulse & its conduction. ● Cardiac cycle: events, pressure & volume changes during different phases ● Heart sounds: types & characteristics ● ECG: definition & interpretations . ● local & humoral control of blood flow ● Cardiac output : definition and factors affecting cardiac output. ● Venous return & Peripheral resistance: definition & factors affecting.. ● Heart rate: definition, normal rate & regulation. ● Radial pulse: definition & characteristics. ● Blood pressure: definition, types, measurement & regulation of arterial blood pressure. Additional /Applied Physiology Cardiac arrhythmias: tachycardia, bradycardia & heart block	L=15 T=12 P=10 IT=01

Respiratory System

Learning Objectives	Contents	Teaching Hours
At the end of the course the students will be able to : ● explain the mechanism of normal respiration ● describe pulmonary volumes and capacities ● summaries the diffusion of gases through the respiratory membrane. ● describe the oxygen & carbon dioxide transport. ● describe the respiratory centers & regulation of respiration. ● define & classify hypoxia and cyanosis.	CORE ● Respiration: definition of external and internal respiration, muscles of respiration and mechanism of respiration. ● Pulmonary and alveolar ventilation, dead space volume ● Pulmonary volumes and capacities ● Composition of atmospheric, alveolar, inspired and expired air. ● Respiratory unit and respiratory membrane. ● Diffusion of Gases through the respiratory membrane. ● Transport of Oxygen & Carbon dioxide in blood & body fluid. ● Nervous & chemical regulation of respiration. ● Lung function tests ● Hypoxia and cyanosis: definition, types ● Definition of Dyspnea, hypercapnea, artificial respiration & Periodic breathing.	L=12 T=10 IT=01

Endocrinology and Reproductive system

Learning Objectives	Contents	Teaching Hours
At the end of the course the students will be able to: ● describe the general mechanism of action of hormone. ● describe the functions & regulation of secretion of individual hormone. ● describe disorders in relation to: ● Pituitary gland ● Thyroid and parathyroid gland ● Adrenal gland ● Endocrine pancreas	CORE : ● Endocrine glands : name & name of their hormones. ● Hormone: definition, classification, mechanism of action. ● Hypothalamic hormones (releasing & inhibitory hormones): name and functions. ● Hormones of pituitary gland: name, functions, regulation of secretion and disorders. ● Hormones of thyroid gland: biosynthesis, transport, functions, regulation of secretion & disorders. ● Hormones of Parathyroid gland: functions, tetany, calcium & phosphate metabolism. ● Hormones of Adrenal gland: name, functions, regulation of secretion & disorders. ● Hormones of Endocrine pancreas : functions, regulation of secretion, consequences of hyperglycaemia and hypoglycaemia	L = 10 hrs T = 8 hrs IT = 1 hrs

Nervous system and special sense

Learning Objectives	Contents	Teaching Hours
At the end of the course the students will be able to: ● explain organization of the nervous system ● explain the basic mechanism of synaptic transmission. ● describe the motor and sensory system of the body. ● describe the functions of cerebellum, thalamus, hypothalamus and basal ganglia ● describe organization & function of autonomic nervous system ● describe the physiology and regulation of body temperature At the end of the course the students will be able to: ● describe the neurophysiology of vision and audition ● describe the physiology of smell and taste	CORE: Functional organization of nervous system and functions of major levels of central nervous system(CNS). Neuron: definition, parts, types Nerve fiber: classification, effects of injury/section to the nerve fiber Synapse: definition and synaptic transmission Sensory receptor, reflex: definition, classification Reflex arc: definition, components Pain: receptor, pathway Neurotransmitter: definition and classification General/somatic senses: definition, classification Name and functions of ascending and descending tracts. Functions of cerebellum, thalamus, hypothalamus, basal ganglia and autonomic nervous system Physiology and regulation of body temperature Special sense (vision, audition, smell and taste): receptor, pathway Additional/ Applied Dual pain pathway, referred pain	L = 14 hrs T = 12 hrs P = 10 hrs

Physiology Practical

Learning Objectives	Contents	Teaching Hours
Students will be able to ● demonstrate the parts of Microscope & skill in using it. ● collect & prepare blood sample. ● determine differential count of WBC. ● estimate haemoglobin. ● determine ESR ● determine Blood grouping (ABO & Rh system) & cross matching. ● determine bleeding time & clotting time. ● interpret Red Cell Indices ● examine the radial pulse & its application. ● measure the blood pressure. ● record the body temperature ● elicit light reflex	CORE: ● Developing skill for use of microscope & common laboratory equipments. ● Collection & preparation of blood sample. ● Determination of differential count of WBC. ● Estimation of haemoglobin. ● Determination of ESR ● Determination of Blood grouping (ABO & Rh system) & cross matching. ● Determination of bleeding time & clotting time. ● Interpretation of Red Cell Indices: PCV, MCV, MCH, MCHC. ● Measurement of Blood pressure ● Examination of radial pulse. ● Recording of the body temperature ● Elicitation of reflex: superficial & deep	40 hr

Biophysics & Biomolecules

Learning Objectives	Contents	Teaching Hours
At the end of the course, students will be able to: ● define biochemistry and explain its importance in medicine. ● define solution, standard solution, colloid and crystalloid . ● define pH, , buffer, acid and base.. ● State the body fluid buffers. ● define and classify isotope. ● state its biomedical importance. ● define and classify carbohydrates, protein & lipid. ● define steroids and sterols. ● define & classify lipoproteins and mention their biomedical importance. ● define and classify enzymes, ● describe the factors affecting enzyme activity. ● define isoenzyme with example and mention their clinical application.	CORE: ● Introduction to Biochemistry ● Concept of solutions ● Colloids and crystalloids. ● Concept of pH and buffer. ● Concept of isotope. ● Concept of biomolecules Carbohydrates. ● Amino acids and proteins. ● Lipids and fatty acids. ● Enzymes	L = 12 hrs T = 15 hrs P = 8 hrs

GIT, Bioenergetics and Metabolism

Learning Objectives	Contents	Teaching Hours
At the end of the course, students will be able to: ● describe the general principles of gastrointestinal function. ● describe the functions of saliva ● demonstrate knowledge about the composition , functions and regulation of secretion of digestive juices. ● explain the functions of liver . ● define digestion, absorption, anabolism and catabolism ● enumerate local hormones of GIT and state their sources and functions. ● describe oxidative phosphorylation. Carbohydrate Metabolism: At the end of the course, students will be able to: ● describe the digestion and absorption of carbohydrate. ● define glycolysis and describe the pathway. ● describe citric acid cycle and explain why it is called an amphibolic and final common metabolic pathway. ● define glycogenesis, glycogenolysis and gluconeogenesis ● describe glucose homeostasis and mention its importance, ● state the glucostatic functions of liver with other biochemical functions.	CORE: ● Local hormones of GIT: name, function & regulation of secretion ● Digestive juices: composition, functions and their regulation of secretion. ● Functions of saliva ● Digestion and absorption of nutrients ● Motor function of GIT: mastication, deglutition, gastric emptying, defecation ● Liver: functions ● Introduction to metabolism ● Oxidative phosphorylation. ● High and low energy compounds. ● Phases of metabolism (digestion, absorption and intermediary metabolism) ● Glycolysis ● Citric acid cycle ● Glycogenesis and glycogenolysis ● Gluconeogenesis ● Blood glucose homeostasis	L = 12 hrs T = 15 hrs IT = 01 hrs

Learning Objectives	Contents	Teaching Hours
Lipid Metabolism At the end of the course, students will be able to: ● describe the digestion and absorption of fat ● enumerate the blood lipids with their sources. ● state the sources and fate of acetyl-CoA. ● State the name, fate and biomedical importance of ketone bodies. ● enumerate the lipoproteins and explain the clinical importance of LDL & HDL cholesterol.	● Digestion and absorption of lipid. ● Blood lipids: Lipid transport and lipoprotein metabolism. ● Ketogenesis and ketosis.	04 hrs
Protein Metabolism At the end of the course, students will be able to: ● describe the digestion and absorption of protein. ● define nitrogen balance, mention its types and state the routes of nitrogen loss. ● define deamination and transamination. ● describe sources and way of disposal of ammonia, explain ammonia intoxication.	CORE: ● Digestion and absorption of protein, ● Nitrogen balance ● Pathways of protein metabolism ● Deamination and transamination. ● Fate of amino acid in the body ● Source and disposal of ammonia ADDITIONAL: ● Role of liver in over all metabolisms.	04 hrs

Food, Nutrition, Vitamins and Minerals

Learning Objectives	Contents	Teaching Hours
At the end of the course, students will be able to: ● define and explain nutrients, essential nutrients, macro and micro minerals, food, proximate principles of food, diet, balanced diet. ● state the full meaning of the abbreviations- BMR, BMI, SDA, RDA, and also define them. ● state the basis of calculating the calorie requirement of a person. ● describe the sources, requirement and function of carbohydrate as nutrient and describe the importance of fibers in diet. ● describe sources, requirement and function of protein as nutrients; mention the name and significance of essential amino acid; ● describe the sources, requirement and function of lipids as nutrients. ● define and classify vitamins. ● describe the sources, function, RDA, deficiency disorders of water soluble vitamins. ● describe the sources, functions, RDA, deficiency disorders and toxicity of fat soluble vitamins. ● state the role of minerals as nutrients, define trace elements. ● state the importance of minerals: sodium, potassium, calcium, iron, iodine, fluoride, selenium, manganese, copper, zinc etc.	CORE: Basic concepts of food, nutrition and dietary principles. Energy balance and calculation of calorie requirement. Nutritional aspect of carbohydrates, fats and proteins, Fibers. Vitamins and minerals.	L = 10 hrs T = 10 hrs

Kidney, body Fluid, electrolyte and acid-base balance

Learning Objectives	Contents	Teaching Hours
At the end of the course the students will be able to: ● mention the functions of kidney ● describe the structure & types of nephron. ● define GFR, plasma load, tubular load, transport maximum, renal threshold & plasma clearance. ● describe the mechanism of urine formation ● describe the body fluid compartments & regulation of body fluid. ● describe the mechanism of acid – base, electrolyte and water balance ● describe the mechanism of acidification of urine and limiting pH of urine. ● state acid base parameters, anion gap and base excess. ● state abnormal constituents in urine(glucose, protein & ketone bodies), normal urine volume and obligatory urine volume. ● define and classify diuresis with example.	CORE: ● Kidney: physiological anatomy & functions ● Nephron: types, parts, structure & functions ● Urine formation: basic mechanism ● GFR: definition, determinants ● Reabsorption and secretion by the renal tubules ● Mechanism of formation of concentrated urine & diluted urine. ● Body fluid: names of body fluid compartment, major anions and cations of ECF & ICF. ● fluid intake and output chart . ● Regulation of normal water balance. ● Major electrolytes and their homeostasis. ● Volume disorders. Additional /Applied Physiology ● plasma clearance, osmolar clearance and free water clearance. ● Acid base homeostasis & disorders.	L = 12 hrs T = 15 hrs IT = 01 hrs

Fundamentals of Molecular Biology and genetics

Learning Objectives	Contents	Teaching Hours
At the end of the course, students will be able to: ● Define nucleic acid, nucleosides, and nucleotides. ● describe the structure and functions of DNA. ● describe the structure, types and functions of RNA. ● define gene, genotype, phenotype. ● define transcription and translation ● mention the importance of medical Biotechnology ● explain the concepts of recombinant DNA technology, PCR	CORE: ● Basic concepts of molecular biology. ● Nucleic acid, nucleosides, and nucleotides. ● transcription and translation. ● Gene, genetic code, mutation. ● PCR, recombinant DNA technology ● importance of medical biotechnology.	L = 05 hrs T = 06 hrs

Clinical Biochemistry and clinical endocrinology

Learning Objectives	Contents	Teaching Hours
At the end of the course, students will be able to: ● state the basic concepts of clinical biochemistry. ● mention the measurements of unit, SI unit ● state the laboratory hazards with its types. ● state the normal level of serum bilirubin and describe the common liver function tests with interpretation. ● explain the basis of application of clinical enzymology in disease. ● state the lipid profiles of blood & their clinical importance. ● state OGTT and its interpretation, define IFG, IGT and HBA1c ● state renal function tests, define proteinuria and microalbuminuria ● state thyroid function tests with interpretation	CORE : ● Introduction to clinical biochemistry. ● Normal biochemical values in conventional and SI. Units. ● Clinical enzymology related to liver and myocardial diseases. ● Lipid profiles and dyslipoproteinemias. ● Organ function tests (liver, kidney & thyroid) ● Diagnosis of diabetes mellitus ● Bilirubin metabolism and Jaundice. ● Proteinuria and microalbuminuria.	L = 6 hrs T = 6 hrs P = 6 hrs

Biochemistry practical

Learning Objectives	Contents	Teaching Hours
Students will be able to: ● list the laboratory hazards and the precautions to prevent them. ● Estimate blood glucose ● Estimate abnormal constituents of urine(sugar, protein & ketone bodies) and their clinical significance. ● interpret the result of: ● Serum urea ● Serum creatinine ● Serum total protein ● Serum bilirubin	CORE ● Estimation of blood glucose ● Estimation of abnormal constituents of urine(sugar, protein & ketone bodies and their clinical significance. ● interpretation of result of: ● Serum urea ● Serum creatinine ● Serum total protein ● Serum bilirubin .	20 hrs

Summative Assessment of Physiology and Biochemistry (First Professional Examination)

Assessment systems and mark distribution

Components	Marks	Total Marks	Contents
WRITTEN EXAMINATION Formative Assessment + MCQ + SAQ	10+20+70 = 100	100	Group – A 1. General Physiology 2. Blood 3. Cardiovascular System 4. Respiratory System 5. Endocrinology & Reproductive system 6. Nervous system & Special sense Group – B 1. Biophysics & Bimolecules 2. Kidney, Body fluid. Electrolyte & Acid-base balance 3. GIT, Bioenergetics & Metabolism 4. Food, Nutrition, Vitamins & Minerals 5. Fundamental of Molecular Biology & Genetics 6. Clinical biochemistry & Clinical Endocrinology
PRACTICAL EXAMINATION OSPE Traditional practical methods and experiments Practical Note Book	40 40 20	100	
ORAL EXAMINATION (SOE)	100	100	
Grand Total		300	

Pass marks 60% in each of written, oral and practical.

Honors Marks 85%

Systems in Term Examination

Term –I Examination	Term –II Examination
1. General Physiology 2. Blood 3. Cardiovascular System 4. Respiratory System 5. Biophysics & Biomolecules 6. GIT, Bioenergetics & Metabolism	1. Kidney, Body fluid. Electrolyte & Acid-base 2. Endocrinology & Reproductive system 3. Nervous system & Special sense 4. Food, Nutrition, Vitamins & Minerals 5. Fundamental of Molecular Biology & Genetics 6. Clinical biochemistry & Clinical Endocrinology

Distribution of Teaching Hours

Physiology (Group-A) 1. Biophysics and Biomolecules

Systems	Lecture hours	Tutorial hours	Practical hours	Integrated teaching hours
1. General Physiology	04	6		
2. Blood	10	10	10	
3. Cardiovascular system	12	10	05	1
4. Respiratory system	10	10		
5. Endocrinology & Reproductive system	12	12		1
6. Nervous system, Special Senses & temperature regulation	12	12	05	
Total	60	60	20	2

Biochemistry (Group B)

Systems	Lecture hours	Tutorial hours	Practical hours	Integrated teaching hours
1. Biophysics and Biomolecules	12	12	8	
2. GIT, Bioenergetics and metabolism	12	12		1
3. Kidney, Body fluid, electrolytes and acid base balance	12	12	6	1
4. Food, nutrition, vitamins and minerals	12	10		
5. Clinical biochemistry and clinical endocrinology	8	8	6	
6. Fundamental of molecular biology and genetics	4	6		
Total	60	60	20	2